

§7.8 - Improper Integrals

Remark (From §2.6)

If $n > 0$

$$\lim_{x \rightarrow \infty} \frac{1}{x^n} = 0$$

and $n < 0$ implies

$$\lim_{x \rightarrow \infty} \frac{1}{x^n} = \infty.$$

While

$$\lim_{x \rightarrow \infty} e^x = \infty \quad \text{and} \quad \lim_{x \rightarrow \infty} e^{-x} = \lim_{x \rightarrow \infty} \frac{1}{e^x} = 0.$$

Definition #1 (Improper Integral I)

If

$$\int_a^t f(x) \, dx$$

exists for all $t \geq a$ then

$$\int_a^\infty f(x) \, dx = \lim_{t \rightarrow \infty} \int_a^t f(x) \, dx$$

is the **improper integral** of f over $[a, \infty)$.

Definition #2 (Improper Integral I)

If

$$\int_t^b f(x) \, dx$$

exists for all $t \leq b$ then

$$\int_{-\infty}^b f(x) \, dx = \lim_{t \rightarrow -\infty} \int_t^b f(x) \, dx$$

is the **improper integral** of f over $(-\infty, b]$.

Example #1

Find

$$\int_0^{\infty} \frac{1}{x^2 + 1} dx.$$

Example #2

Find

$$\int_2^{\infty} \frac{1}{x \ln(x)} dx.$$

Definition #3

Suppose

$$I = \int_a^\infty f(x) \, dx \in \mathbf{R}.$$

Then the improper integral **converges**.

If $I \notin \mathbf{R}$, then the improper integral **diverges**.

Warning

We can have

$$\lim_{x \rightarrow \infty} f(x) = 0 \quad \text{and} \quad \int_0^{\infty} f(x) \, dx = \infty.$$

Example #3

Find

$$\int_1^{\infty} \frac{1}{x} dx.$$

Definition #4 (Improper Integral I)

If

$$I_1 = \int_{-\infty}^a f(x) \, dx \quad \text{and} \quad I_2 = \int_a^{\infty} f(x) \, dx$$

converge then

$$\int_{\mathbf{R}} f(x) \, dx = \int_{-\infty}^{\infty} f(x) \, dx = I_1 + I_2.$$

Warning

The terms

$$\int_{\mathbf{R}} f(x) \, dx \quad \text{and} \quad \lim_{t \rightarrow \infty} \int_{-t}^t f(x) \, dx$$

need **not** be equal.

Example #4

Find

$$\int_{\mathbf{R}} \sin(x) \, dx.$$

Definition #5 (Improper Integral II)

Let f be a continuous function on $[a, b)$ with

$$\lim_{x \rightarrow b^-} f(x) = \pm\infty.$$

Then

$$\int_a^b f(x) \, dx = \lim_{t \rightarrow b^-} \int_a^t f(x) \, dx$$

is the **improper integral** of f over $[a, b]$.

Definition #6 (Improper Integral II)

Let f be a continuous function on $(a, b]$ with

$$\lim_{x \rightarrow a^+} f(x) = \pm\infty.$$

Then

$$\int_a^b f(x) \, dx = \lim_{t \rightarrow a^+} \int_t^b f(x) \, dx$$

is the **improper integral** of f over $[a, b]$.

Definition #7 (Improper Integral II)

Let f be a continuous function on $[a, c) \cup (c, b]$ with f unbounded at $c \in (a, b)$.

If

$$I_1 = \int_a^c f(x) \, dx \quad \text{and} \quad I_2 = \int_c^b f(x) \, dx$$

converge then the **improper integral**

$$\int_a^b f(x) \, dx = I_1 + I_2$$

converges.

Example #5

Find

$$\int_{-1/2}^1 \frac{1}{\sqrt{2x+1}} dx.$$

Example #6

Find

$$\int_0^{\pi/2} \tan(t) \, dt.$$

Example #7 (Two Types)

Find

$$\int_{\mathbf{R}} \frac{1}{x^2} dx.$$

Theorem #1 (Comparison Test)

Suppose f and g are continuous on the interval $[a, \infty)$ where

$$0 \leq g(x) \leq f(x)$$

for all $x \geq a$.

Then

1. If $\int_a^\infty f(x) \, dx$ converges then $\int_a^\infty g(x) \, dx$ converges.
2. If $\int_a^\infty g(x) \, dx$ diverges then $\int_a^\infty f(x) \, dx$ diverges.

Example #8

One can show:

1. $\int_1^\infty 1/x^p \, dx$ converges if and only if $p > 1$.
2. $\int_0^1 1/x^p \, dx$ converges if and only if $p < 1$.
3. $\int_0^\infty 1/b^x \, dx$ converges for $b > 1$.
4. $\int_{-\infty}^0 b^x \, dx$ converges for $b > 1$.

You need to prove these on the exams.

Example #9

Find

$$\int_1^{\infty} \frac{\ln^2(x)}{x} dx$$

Example #10

Find

$$\int_1^{\infty} \frac{\sin(x) + 2}{x^2} dx.$$

Example #11

Find

$$\int_1^{\infty} \frac{\ln(x)}{x^2} dx$$

Example #12

Find the volume of the solid revolution of the infinitely large region in quadrant I bounded by

$$x = 1, \quad \text{and} \quad y = \frac{1}{x}, \quad \text{and} \quad y = 0$$

revolved around the x -axis.

Example #13

Determine which is larger:

$$\int_0^{\infty} x^2 e^{-x^2} dx \quad \text{or} \quad \frac{1}{2} \int_0^{\infty} e^{-x^2} dx$$